

Nmap Learning Guide

4-Week Daily Practice Sessions — The Config Bench Lab

Primary Workstation: **192.168.0.11** (Ubuntu)

Router / Gateway: **192.168.0.1**

Secondary Device: **192.168.0.197** (Apple — MDNS detected)

Subnet: **192.168.0.0/24**

Results folder: **~/nmap_results/** — create it first: `mkdir -p ~/nmap_results`

■■ GOLDEN RULE: Only scan networks and devices you own or have explicit written permission to scan. Unauthorized scanning is illegal. Home lab = fair game. Credit union = written authorization from management required first.

WEEK 1 — Discovery & Basics

Monday — Your First Network Map

Goal: See every device on your network

```
sudo nmap -sn 192.168.0.0/24
```

What to look for:

- How many hosts are up?
- Do you recognize every device?
- Note the MAC addresses — they reveal the manufacturer
- Compare to what you EXPECT to see

Save your results:

```
sudo nmap -sn 192.168.0.0/24 -oN ~/nmap_results/week1_monday.txt
```

Tuesday — Port Scanning Your Own Box

Goal: See what services your Ubuntu workstation is exposing

```
sudo nmap 192.168.0.11
```

```
sudo nmap -v 192.168.0.11
```

- Which ports are OPEN?
- Which are CLOSED vs FILTERED?
- Any ports open you didn't expect?
- Port 22 (SSH)? Port 80/443 (web server)?

Wednesday — Service & Version Detection

Goal: Find out WHAT is running on each open port

```
sudo nmap -sV 192.168.0.11
```

Example output: 22/tcp open ssh OpenSSH 8.9p1 Ubuntu Version numbers matter — outdated versions = potential vulnerabilities.

Thursday — OS Detection

Goal: Let Nmap fingerprint the operating system

```
sudo nmap -O 192.168.0.11
```

```
sudo nmap -O 192.168.0.1 # Try your router too
```

- Does it correctly identify Ubuntu?
- What does it think your router OS is?
- "OS details" line is the key result

Friday — Week 1 Review Scan

Goal: Run a combined scan and save a full report

```
sudo nmap -A 192.168.0.0/24 -oN ~/nmap_results/week1_full_scan.txt
```

-A flag includes: OS detection + Service/version detection + Script scanning + Traceroute

WEEK 2 — Scan Types & Stealth

Monday — TCP SYN Scan (Stealth Scan)

Goal: Understand the default scan method — and see it in Wireshark

```
sudo nmap -sS 192.168.0.11
```

■ Wireshark Lab Exercise: Run Wireshark simultaneously and apply this filter: `tcp.flags.syn == 1 && tcp.flags.ack == 0`
Watch the SYN packets fly out in real time. This is the scan's fingerprint.

- Sends SYN → waits for SYN-ACK → sends RST (never completes handshake)
- Less likely to appear in application logs
- This IS the default when running as root

Tuesday — TCP Connect Scan

Goal: Compare full connect scan vs SYN scan in Wireshark

```
sudo nmap -sT 192.168.0.11
```

- Completes the full TCP 3-way handshake
- MORE visible in logs than SYN scan
- Run Wireshark simultaneously — compare packet patterns to Monday's scan

Wednesday — UDP Scan

Goal: Don't forget UDP — many critical services hide there

```
sudo nmap -sU 192.168.0.11
```

■ **Warning: This is SLOW — UDP has no handshake so Nmap waits for timeouts.**

- 53 — DNS
- 67/68 — DHCP
- 123 — NTP
- 161 — SNMP (critical for network admins!)
- 500 — IPSec VPN

Thursday — Specific Port Scanning

Goal: Target specific ports instead of the default top 1000

```
sudo nmap -p 22,80,443,8080,8443 192.168.0.11
```

```
sudo nmap -p 1-1024 192.168.0.11
```

```
sudo nmap -p- 192.168.0.11 # ALL 65535 ports (slow but thorough)
```

Credit Union Note: Audits often require full port scans. Default Nmap only hits the top 1000. Attackers scan ALL 65535 — you should know what they'd find on your network.

Friday — Scan the Router

Goal: Audit your gateway device

```
sudo nmap -A 192.168.0.1
```

- Is the admin interface exposed? (port 80/443/8080)
- Is SSH open on the router?
- Any unexpected services?
- Note firmware/version if detected

WEEK 3 — NSE Scripts (Nmap Scripting Engine)

Nmap has 600+ built-in scripts. NSE scripts extend Nmap from a port scanner into a full reconnaissance and vulnerability detection platform.

Monday — Introduction to NSE

```
ls /usr/share/nmap/scripts/ | head -30 # Browse available scripts
sudo nmap -sC 192.168.0.11 # Run default scripts
```

- **auth** — Authentication bypass tests
- **vuln** — Vulnerability detection
- **discovery** — Service enumeration
- **brute** — Brute force — use carefully and only on your own devices

Tuesday — HTTP Enumeration

Goal: Find every web interface on your network

```
sudo nmap -p 80,443,8080,8443 --script http-title,http-server-header 192.168.0.0/24
```

Finds web interfaces on routers, printers, cameras, NAS devices — anything with an HTTP server.

Wednesday — SMB / Windows Scanning

Goal: Highly relevant for your credit union environment

```
sudo nmap -p 445 --script smb-os-discovery 192.168.0.0/24
sudo nmap -p 445 --script smb-vuln-ms17-010 192.168.0.0/24
```

■■ EternalBlue (MS17-010) is the exploit used in WannaCry ransomware. Still found in unpatched systems in 2026. If it shows vulnerable — that is a critical finding.

Thursday — Vulnerability Scan (Your Own Box Only)

Goal: Run a vuln scan against yourself — see what an attacker would see

```
sudo nmap --script vuln 192.168.0.11
```

■■ Only run **--script vuln** against devices **YOU** own and control.

Friday — Save Professional Output

Goal: Generate multi-format reports

```
sudo nmap -A 192.168.0.0/24 \
-oN ~/nmap_results/week3_normal.txt \
-oX ~/nmap_results/week3_xml.xml \
-oG ~/nmap_results/week3_grep.txt
```

■ Upload the results file to Fred and I'll generate a full professional PDF report — same as your Wireshark capture report. Just paste the output or attach the file.

WEEK 4 — Real World Application

Monday — Baseline Your Network

Goal: Create a documented monthly baseline

```
sudo nmap -A 192.168.0.0/24 -oN ~/nmap_results/baseline_$(date +%Y%m%d).txt
```

Run this every month. New device appeared? Investigate. New port open that wasn't before? Investigate. This IS network security.

Tuesday — Timing & Performance

Goal: Understand scan speed vs stealth tradeoff

```
sudo nmap -T1 192.168.0.11 # Sneaky (very slow)
sudo nmap -T3 192.168.0.11 # Normal (default)
sudo nmap -T4 192.168.0.11 # Aggressive (good for your own lab)
```

■ Wireshark Lab: Run T1 then T4 with Wireshark open. The packet timing difference is dramatic and directly shows why timing matters for IDS evasion.

Wednesday — Firewall Evasion Techniques

Goal: Understand attacker techniques so you can detect them

```
sudo nmap -f 192.168.0.11 # Fragment packets
sudo nmap -D RND:5 192.168.0.11 # Decoy scan
sudo nmap --source-port 53 192.168.0.11 # Spoof source port
```

These are defensive awareness exercises. Run Wireshark simultaneously on each — seeing what they look like on the wire is the actual lesson.

Thursday — SNMP Scanning (Critical for Network Admins)

Goal: SNMP is everywhere in enterprise networks — and often misconfigured

```
sudo nmap -sU -p 161 192.168.0.0/24
sudo nmap -sU -p 161 --script snmp-brute 192.168.0.0/24
sudo nmap -sU -p 161 --script snmp-info 192.168.0.0/24
```

Credit Union Alert: SNMP v1/v2 uses CLEARTEXT community strings. If you find "public" or "private" still set on any device — that is an immediate finding. SNMP v3 with authentication is the standard you should be enforcing.

Friday — Your First Audit Report

Goal: Produce a complete scan and generate a professional report with Fred

```
mkdir -p ~/nmap_results
sudo nmap -A -p- --open 192.168.0.0/24 \
-oX ~/nmap_results/audit_$(date +%Y%m%d).xml
```

Upload the XML or paste the text output to Fred → I'll generate a full PDF report with findings, risk ratings, and recommendations. Same workflow as your Wireshark packet analysis.

Quick Reference Card

Command	What It Does
<code>nmap -sn 192.168.0.0/24</code>	Ping scan – discover hosts only
<code>nmap 192.168.0.11</code>	Default port scan (top 1000 ports)
<code>nmap -sV 192.168.0.11</code>	Service & version detection
<code>nmap -O 192.168.0.11</code>	OS fingerprinting
<code>nmap -A 192.168.0.11</code>	All of the above + scripts + traceroute
<code>nmap -sS 192.168.0.11</code>	SYN stealth scan (default as root)
<code>nmap -sT 192.168.0.11</code>	Full TCP connect scan
<code>nmap -sU 192.168.0.11</code>	UDP scan (slow)
<code>nmap -p- 192.168.0.11</code>	Scan all 65,535 ports
<code>nmap -p 22,80,443</code>	Scan specific ports only
<code>nmap --script vuln</code>	Run vulnerability detection scripts
<code>nmap --script smb-vuln-ms17-010</code>	Check for EternalBlue/WannaCry
<code>nmap -sU -p 161 --script snmp-info</code>	SNMP enumeration
<code>nmap -T4 192.168.0.11</code>	Aggressive timing (faster scan)
<code>nmap -oN file.txt</code>	Save output – normal text format
<code>nmap -oX file.xml</code>	Save output – XML format (importable)
<code>nmap -oG file.txt</code>	Save output – greppable format